

# Project 3

EECS 116/216 Fall 2026

UC Berkeley Engineering

BAIR

BERKELEY ARTIFICIAL INTELLIGENCE RESEARCH

ARPL Agile Robotics and Perception Lab

https://arplaboratory.github.io/

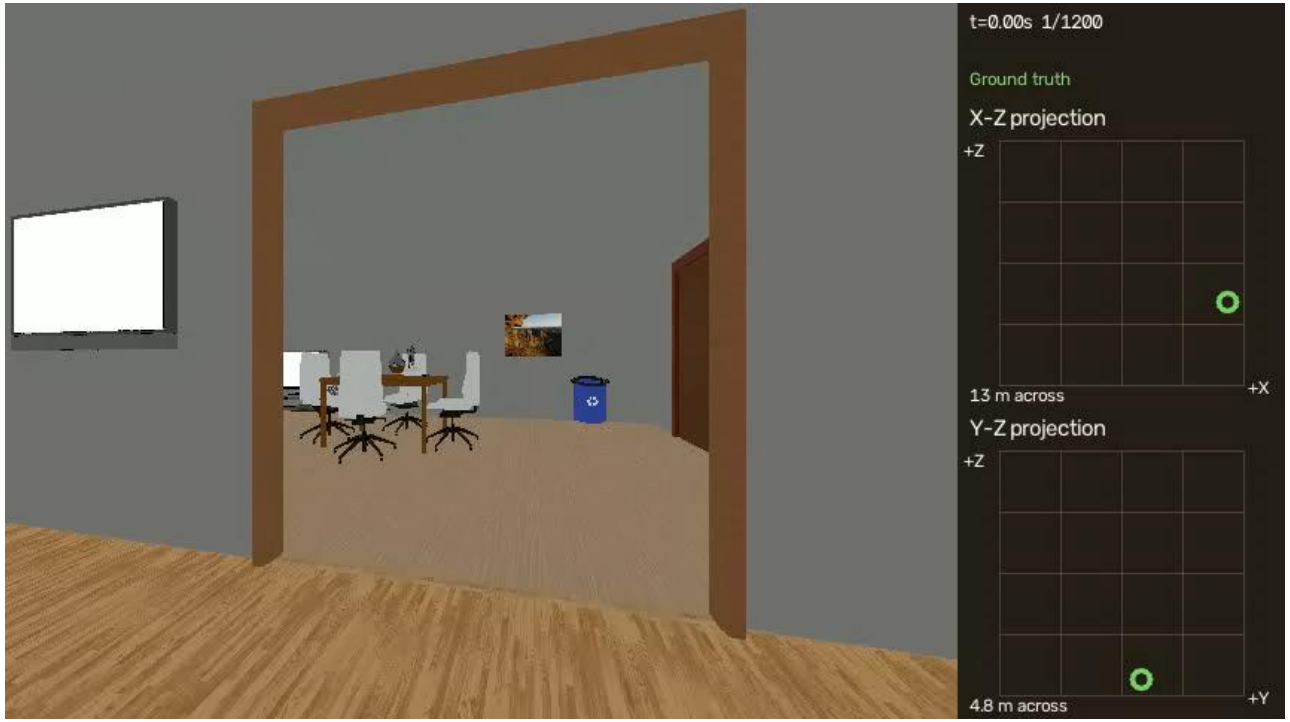
# Teams

- Groups of 3
  - At least one undergrad (enrolled in EECS 116)
  - At least one grad (enrolled in EECS 216)
- Sign up form: https://forms.gle/6rLpqzZswP4pFqAL8
  - Link in bCourses announcement
  - Open until today 11:59pm
  - Afterwards we will manually assign teams

# Challenge: Compute-Aware VIO

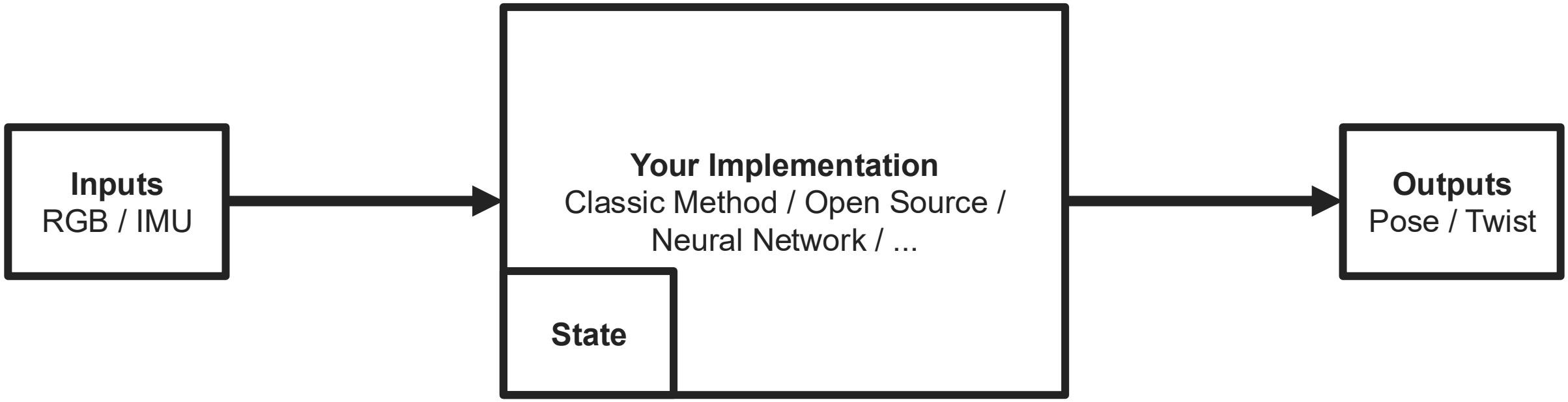
**Task**: Implementing a compute efficient VIO system

- Input
  - RGB frames 320x240/640x480 (20Hz)
  - IMU (acceleration, angular rates) (200Hz)
- Output:
  - Pose (relative to first frame, metric)
    - Position
    - Orientation
  - Twist
    - Linear velocity
    - Angular velocity

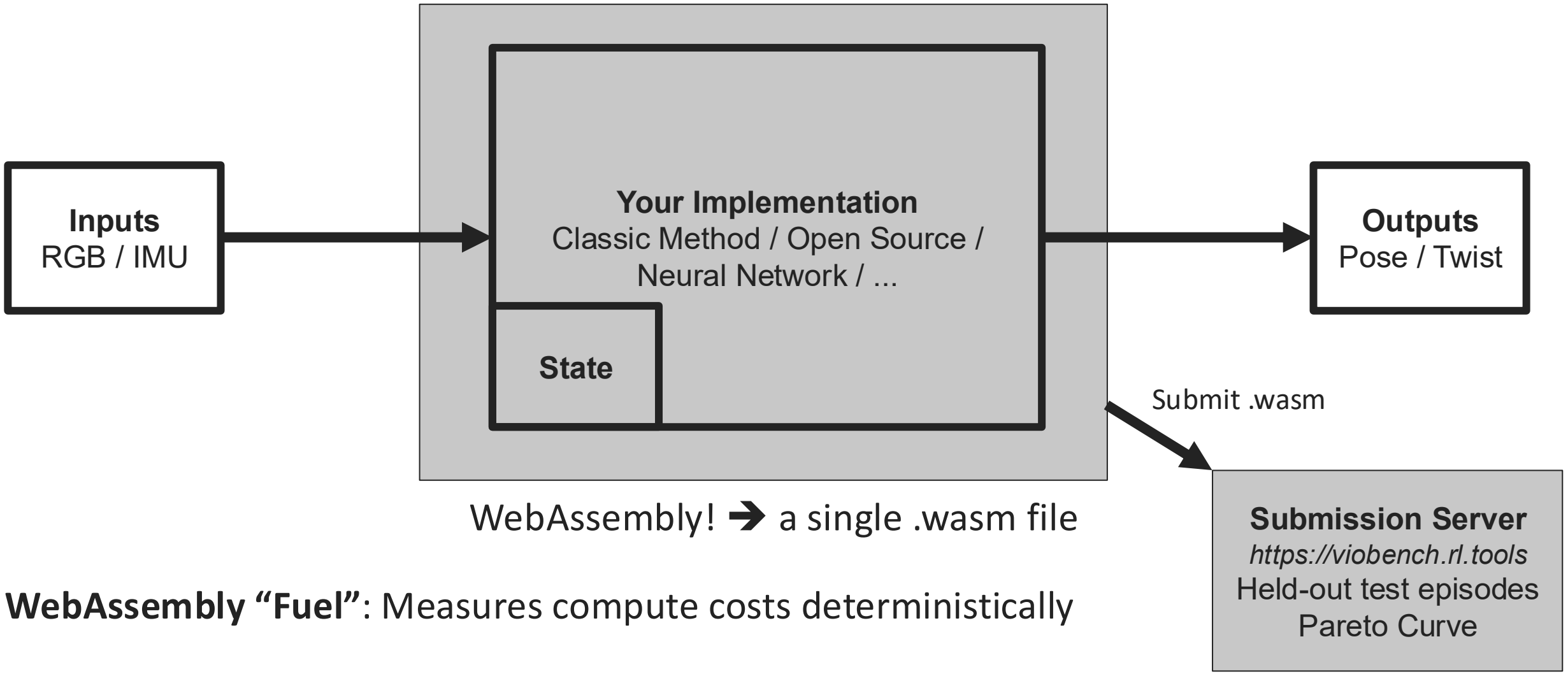


# Methods

Up to you!



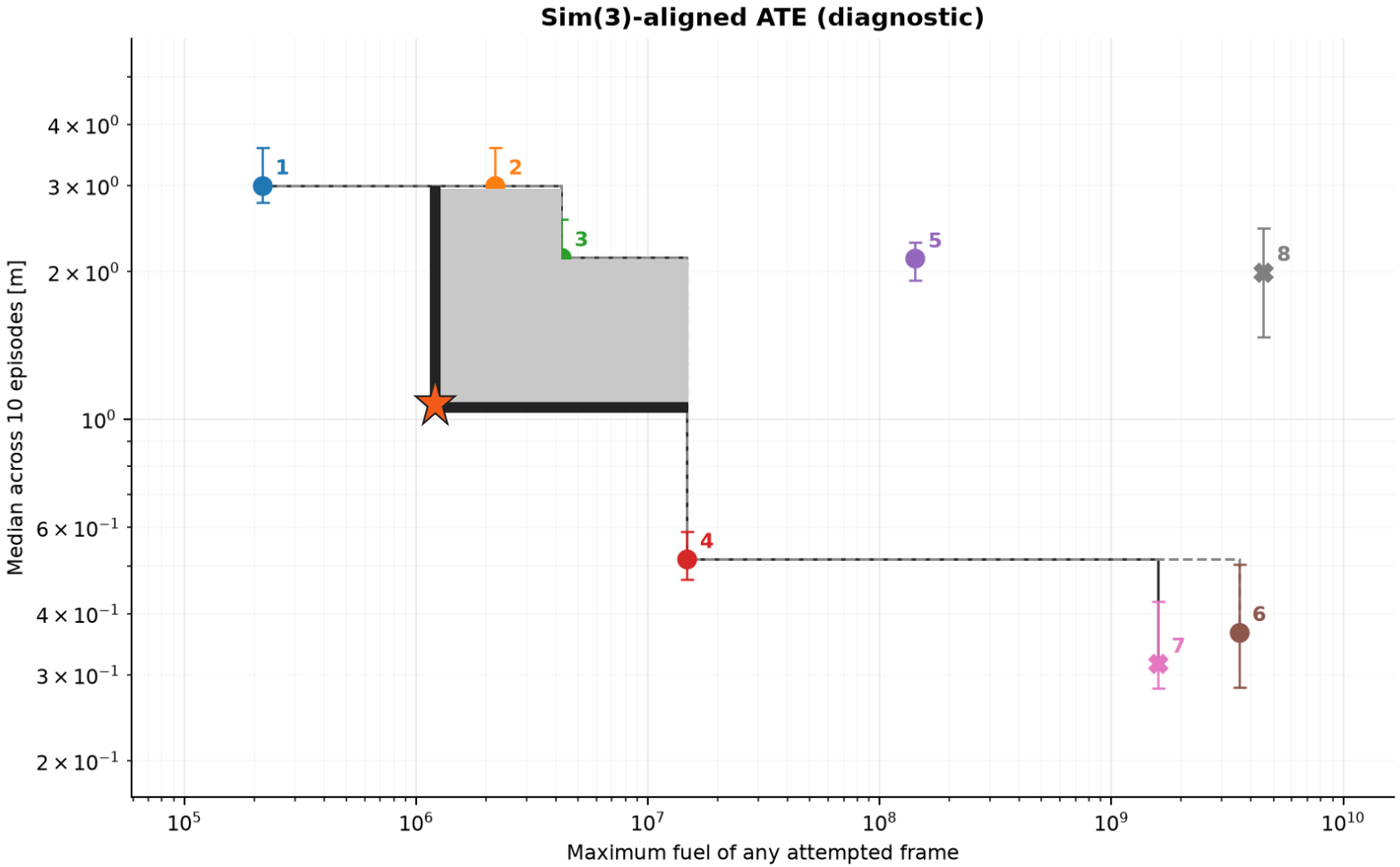
# Compute-Awareness




WebAssembly! ➔ a single .wasm file

**WebAssembly "Fuel"**: Measures compute costs deterministically

# Pareto Curve



# Compute & Submissions

**Compute**

- Two teams per lab workstation (RTX 5080)
- Alternating daily schedule (handover at noon)

**Submissions**

- https://viobench.rl.tools
- Group identified by token (sent to the emails registered in the form)
- One per day per team

# AI Policy

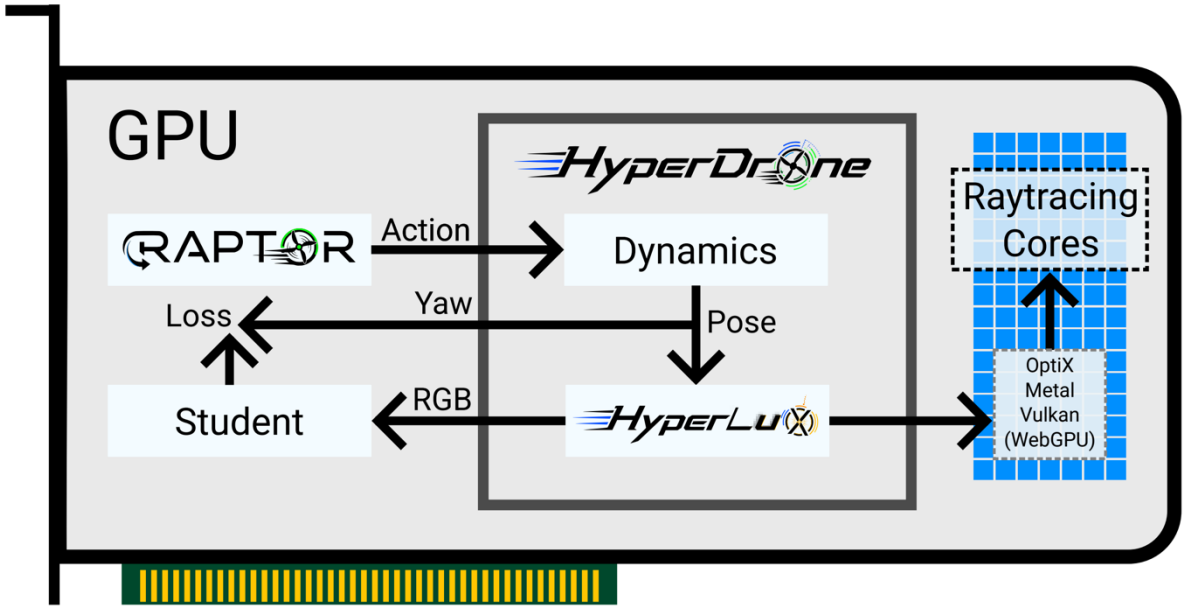
**“Everything Goes!”**

- Industry & Academia now heavily rely on agentic coding
  - ➔ Realistic project workflow
- **Only requirement**: Your solution needs to be reproducible
  - E.g. if you rely on ML, your training pipeline needs to be repeatable
- Check out the Google Student Plan
  - Free for one year
  - You can use Antigravity (CLI)
- Non AI-based, artisanal solutions are also appreciated
  - Robustness, simplicity of the architecture etc.
  - Please highlight this in the report

# Learning-Based Methods

- Should be based on **HyperDrone** (for data-generation)
- Example
  - Included in the viobench package
  - Training a CNN using HyperDrone
  - Bundling the trained model into a .wasm





# Quickstart

viobench

```
> pip install viobench
> viobench init python my_solution
> viobench bundle my_solution/solution -o my_solution.wasm
```

Documentation & Code

viobench

```
> git clone https://github.com/jonas-eschmann/viobench
```

# Training on Lab Workstations

**Conda:**

/share/instsww/anaconda3/bin/conda init

Restart terminal

conda config --set auto_activate_base false

conda deactivate

conda create -n myenv python=3.14 -c conda-forge

conda activate myenv

conda install -c nvidia/label/cuda-13.2.0 "cuda-toolkit=13.2" "cuda-nvcc=13.2" "cuda-version=13.2"

conda install cmake

**Train and bundle model**

git clone https://github.com/jonas-eschmann/viobench

pip install -e viobench[train]

cd viobench/viobench/solution/examples/python/tiny_cnn

python train.py --generate 8 --epochs 10

python3 -m viobench bundle solution -o tiny_cnn.wasm

python3 -m viobench check tiny_cnn.wasm

# Timeline (Tentative)

| Week | |
|---|---|
| 11/02 | Soft-launch |
| 11/09 | First stage metrics and test datasets finalized; Pareto frontier based on examples/baselines |
| 11/16 | First stage progress presentations; New Pareto frontier frozen; (New metrics / test datasets) |
| 11/23 | Second stage progress presentations; New Pareto frontier frozen; (New metrics / test datasets) |
| 11/30 | Third and final presentations; Final Pareto frontier |
| TBD | Report submission deadline |

# Progress Presentation Attendance

- One miss permitted (in either progress presenation 1 or 2)
  - But at least someone from the team needs to present
- Final presentations: attendance mandatory for all